ЛР 2. Программа в отладчике

Система: macOS, ARM64. Отладчик: LLDB в VS Code.

1. Без отладчика программа вывела agent: total=6 area=12.

2. Точка останова установлена на строке 15.

3. До строки 15 отладчик показал p.x = -16, p.y = -1, total = 1. После присваивания p.x = 3 и p.y = 4. Объявление массива на строке 16 не выполняет отдельную команду.

4. После strcpy в name[0] записан код 97 (буква a). Завершающий ноль находится в name[5].

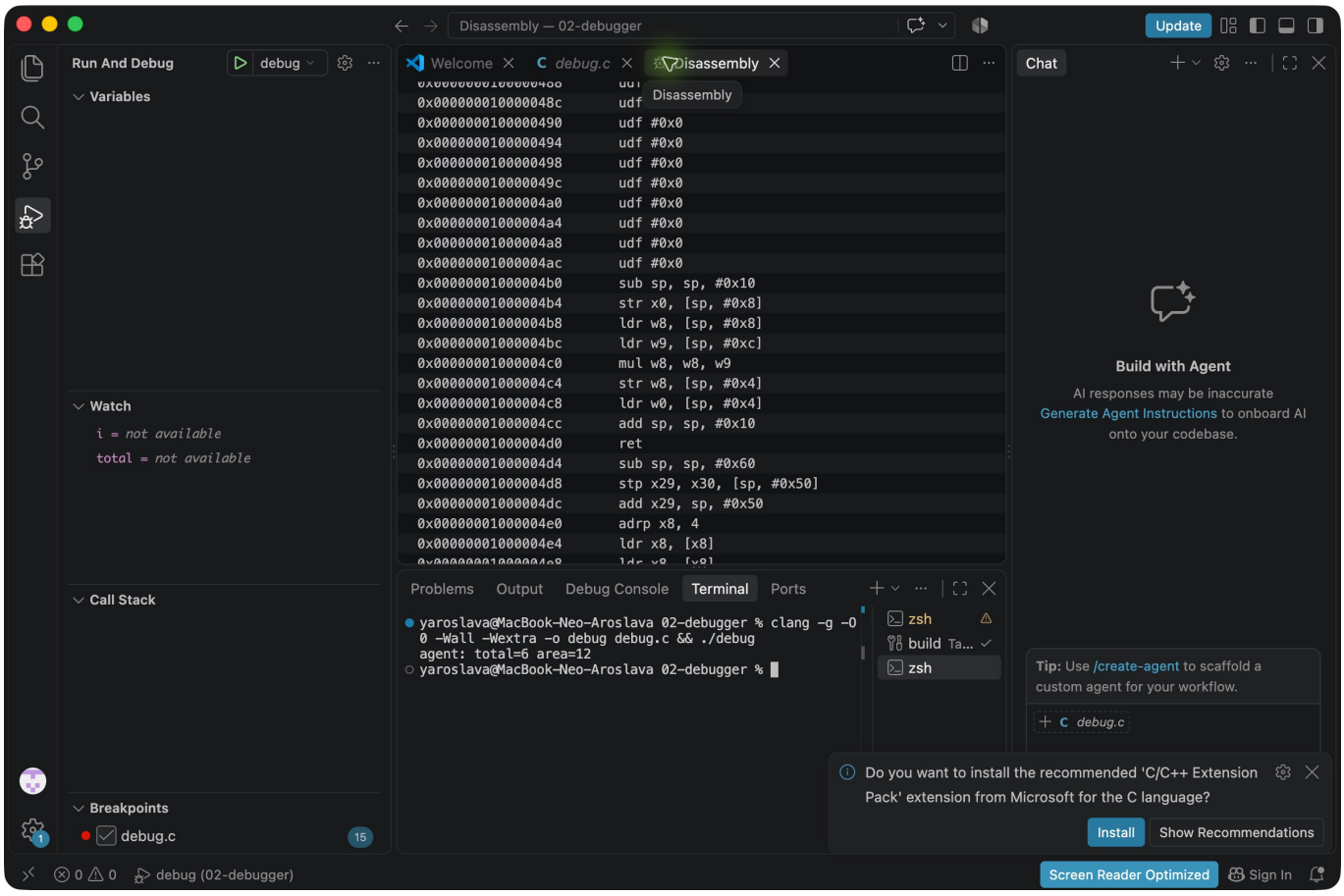
5. На втором проходе i = 2 и total = 1. После прибавления total = 3.

6. В стеке три кадра: area, main и системный start.

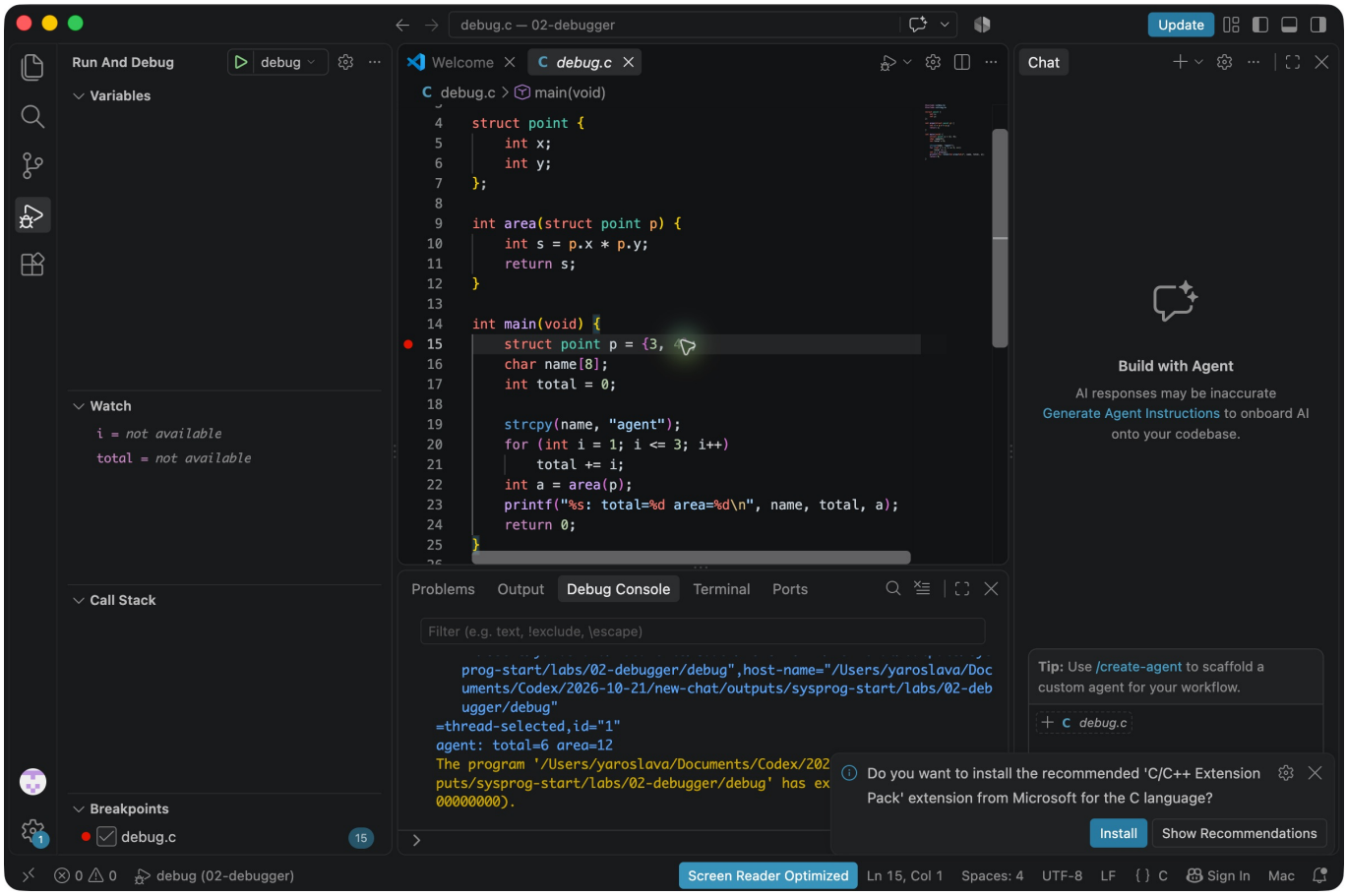
7. Команда умножения: mul w8, w8, w9. Перед printf значения a = 12 и total = 6.

8. Под отладчиком программа вывела agent: total=6 area=12 и завершилась с кодом 0.

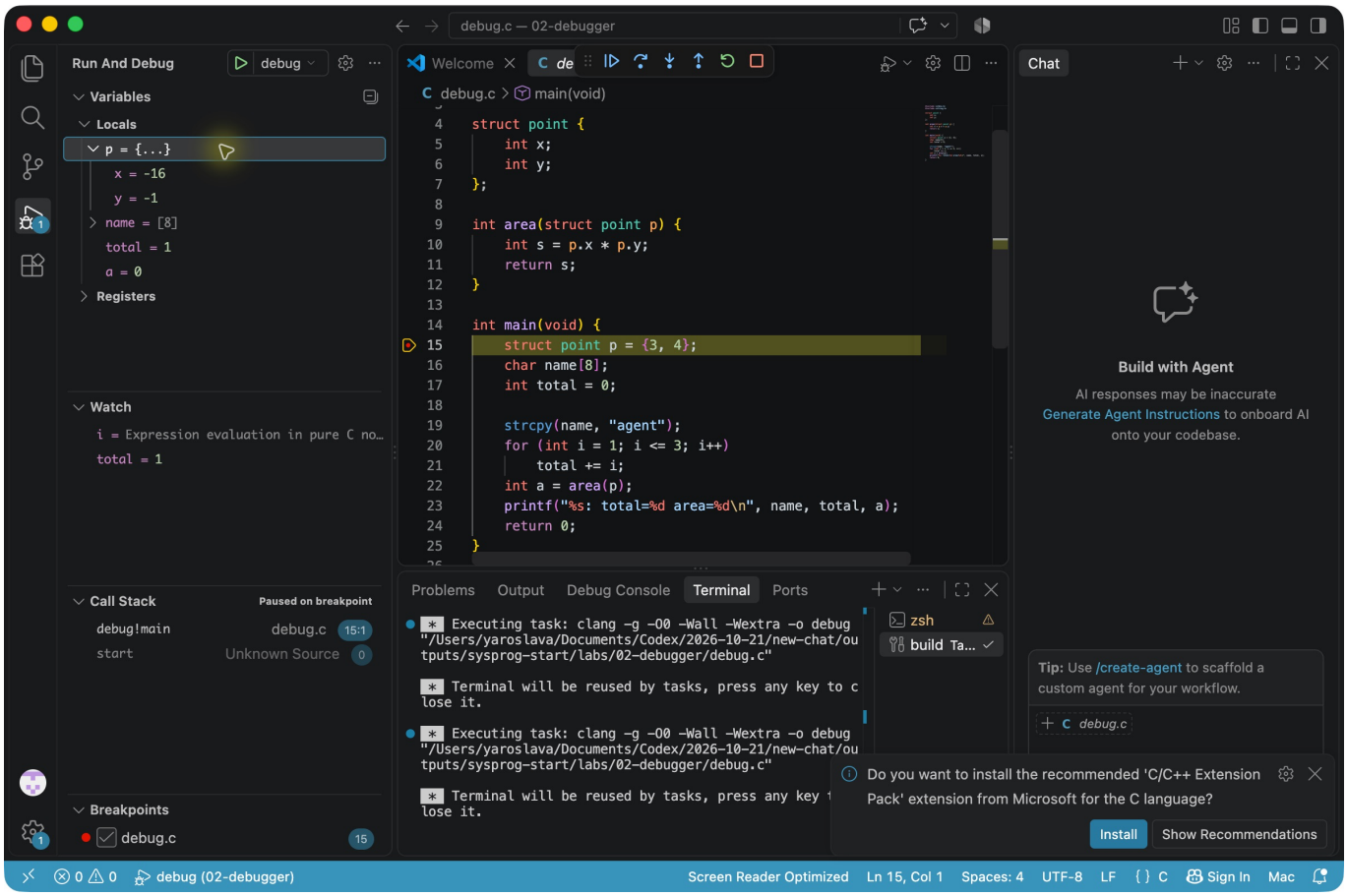
1



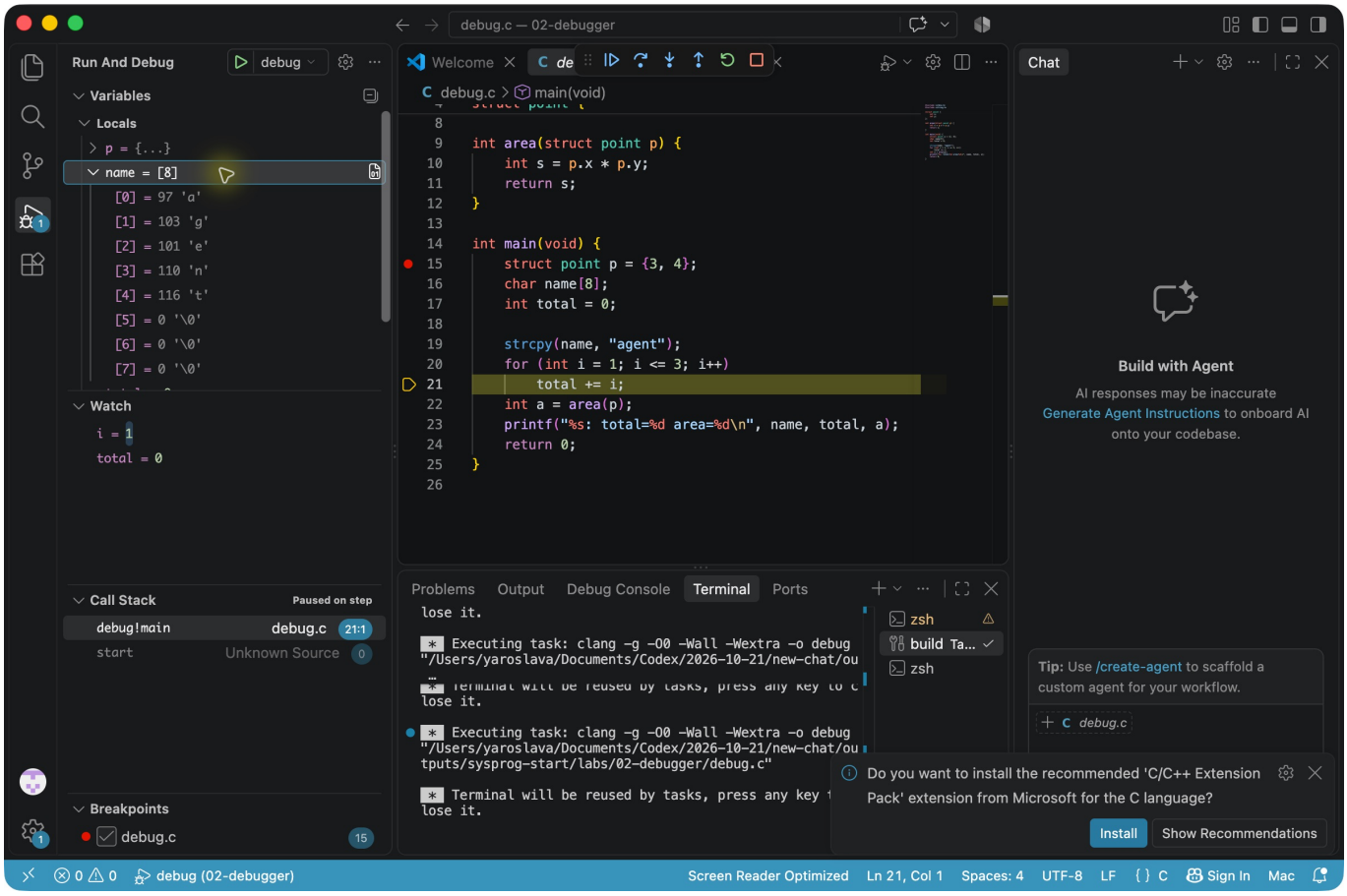
2


debug.c — 02-debugger
Update
Run And Debug
debug
Variables
Watch
i = not available
total = not available
Call Stack
Breakpoints
debug.c
15
Welcome
debug.c
debug.c > main(void)
struct point {
int x;
int y;
};
int area(struct point p) {
int s = p.x * p.y;
return s;
}
int main(void) {
struct point p = {3,
char name[8];
int total = 0;
strcpy(name, "agent");
for (int i = 1; i <= 3; i++)
total += i;
int a = area(p);
printf("%s: total=%d area=%d\n", name, total, a);
return 0;
}
Problems
Output
Debug Console
Terminal
Ports
Filter (e.g. text, !exclude, \escape)
prog-start/labs/02-debugger/debug",host-name="/Users/yaroslava/Documents/Codex/2026-10-21/new-chat/outputs/sysprog-start/labs/02-debugger/debug"
=thread-selected,id="1"
agent: total=6 area=12
The program '/Users/yaroslava/Documents/Codex/2026-10-21/new-chat/outputs/sysprog-start/labs/02-debugger/debug' has exited with code 0 (0x00000000).
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debug.c
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debug (02-debugger)
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Spaces: 4
UTF-8
LF
C
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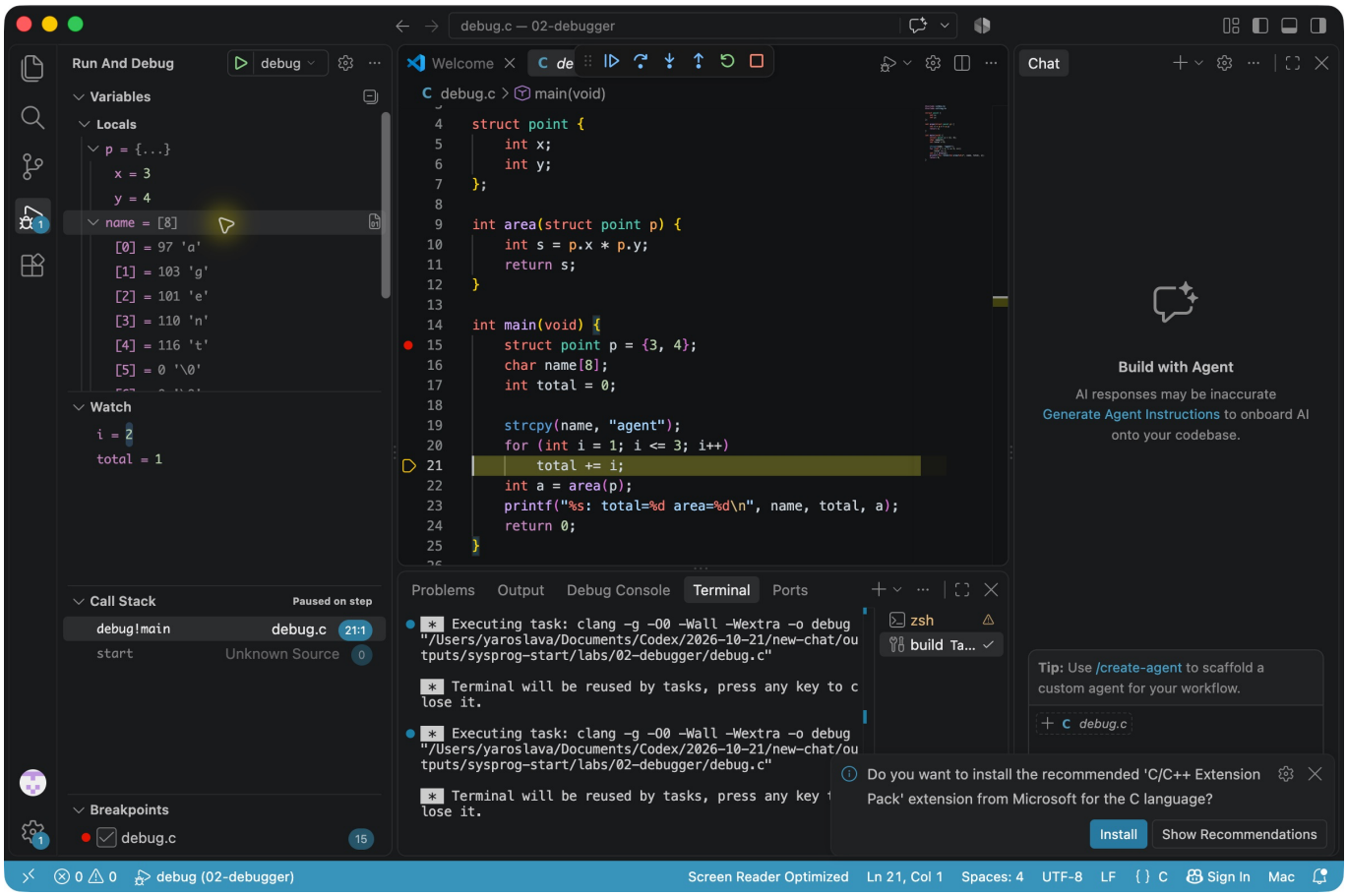
3


debug.c — 02-debugger
Run And Debug
debug
Variables
Locals
p = {...}
x = -16
y = -1
name = [8]
total = 1
a = 0
Registers
Watch
i = Expression evaluation in pure C no…
total = 1
Call Stack
Paused on breakpoint
debug!main
debug.c
15:1
start
Unknown Source
0
Breakpoints
debug.c
15
Welcome
debug.c
debug.c > main(void)
struct point {
int x;
int y;
};
int area(struct point p) {
int s = p.x * p.y;
return s;
}
int main(void) {
struct point p = {3, 4};
char name[8];
int total = 0;
strcpy(name, "agent");
for (int i = 1; i <= 3; i++)
total += i;
int a = area(p);
printf("%s: total=%d area=%d\n", name, total, a);
return 0;
}
Problems
Output
Debug Console
Terminal
Ports
* Executing task: clang -g -O0 -Wall -Wextra -o debug "/Users/yaroslava/Documents/Codex/2026-10-21/new-chat/outputs/sysprog-start/labs/02-debugger/debug.c"
* Terminal will be reused by tasks, press any key to close it.
* Executing task: clang -g -O0 -Wall -Wextra -o debug "/Users/yaroslava/Documents/Codex/2026-10-21/new-chat/outputs/sysprog-start/labs/02-debugger/debug.c"
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zsh
build Ta…
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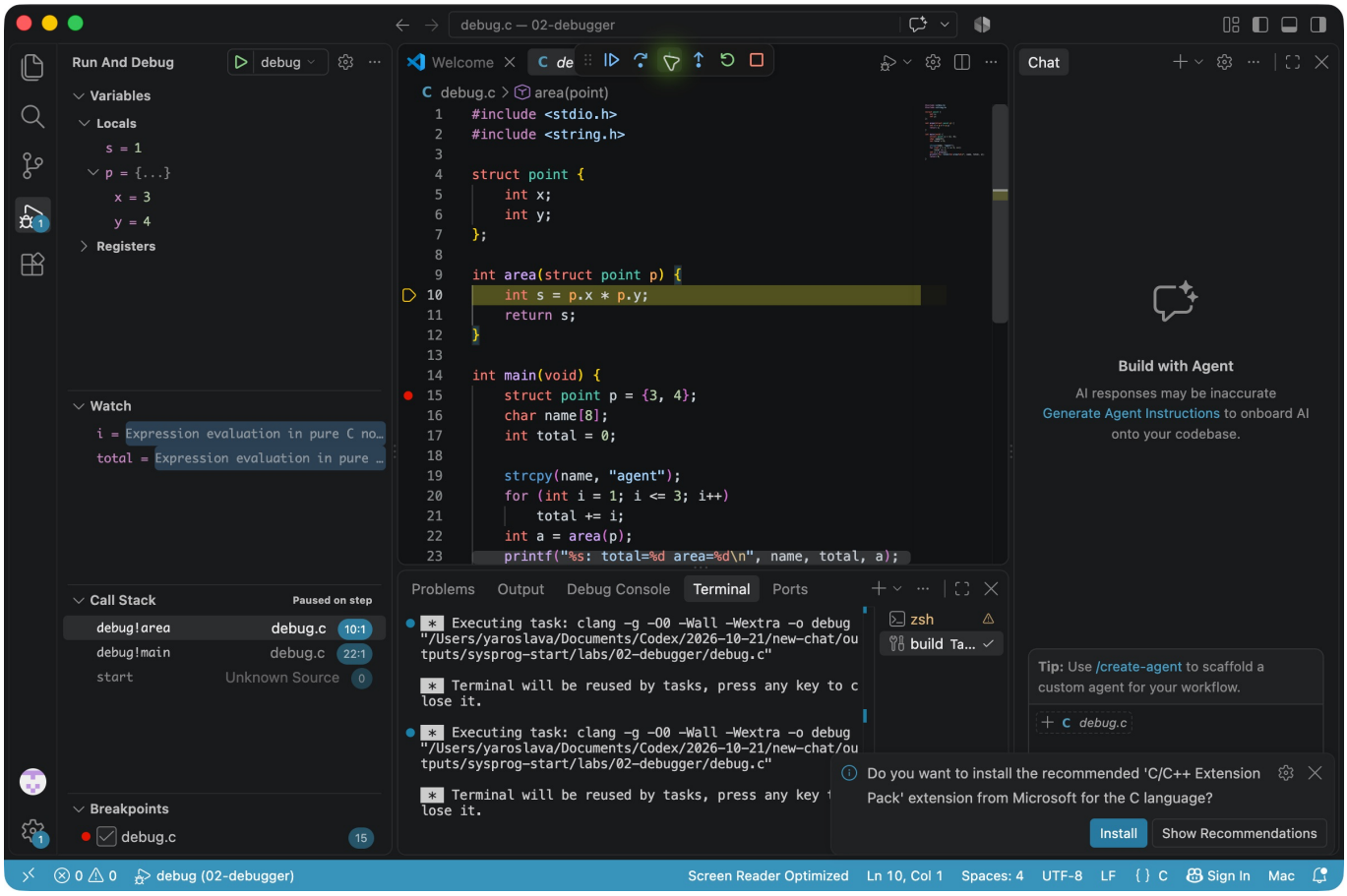
4


debug.c — 02-debugger
Run And Debug
debug
Variables
Locals
p = {...}
name = [8]
[0] = 97 'a'
[1] = 103 'g'
[2] = 101 'e'
[3] = 110 'n'
[4] = 116 't'
[5] = 0 '\0'
[6] = 0 '\0'
[7] = 0 '\0'
Watch
i = 1
total = 0
Call Stack
Paused on step
debug!main
debug.c 21:1
start
Unknown Source 0
Breakpoints
debug.c 15
Welcome
debug.c > main(void)
8
9 int area(struct point p) {
10 int s = p.x * p.y;
11 return s;
12 }
13
14 int main(void) {
15 struct point p = {3, 4};
16 char name[8];
17 int total = 0;
18
19 strcpy(name, "agent");
20 for (int i = 1; i <= 3; i++)
21 total += i;
22 int a = area(p);
23 printf("%s: total=%d area=%d\n", name, total, a);
24 return 0;
25 }
26
Problems
Output
Debug Console
Terminal
Ports
lose it.
Executing task: clang -g -O0 -Wall -Wextra -o debug "/Users/yaroslava/Documents/Codex/2026-10-21/new-chat/ou
Terminal will be reused by tasks, press any key to c
lose it.
Executing task: clang -g -O0 -Wall -Wextra -o debug "/Users/yaroslava/Documents/Codex/2026-10-21/new-chat/ou
tputs/sysprog-start/labs/02-debugger/debug.c"
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lose it.
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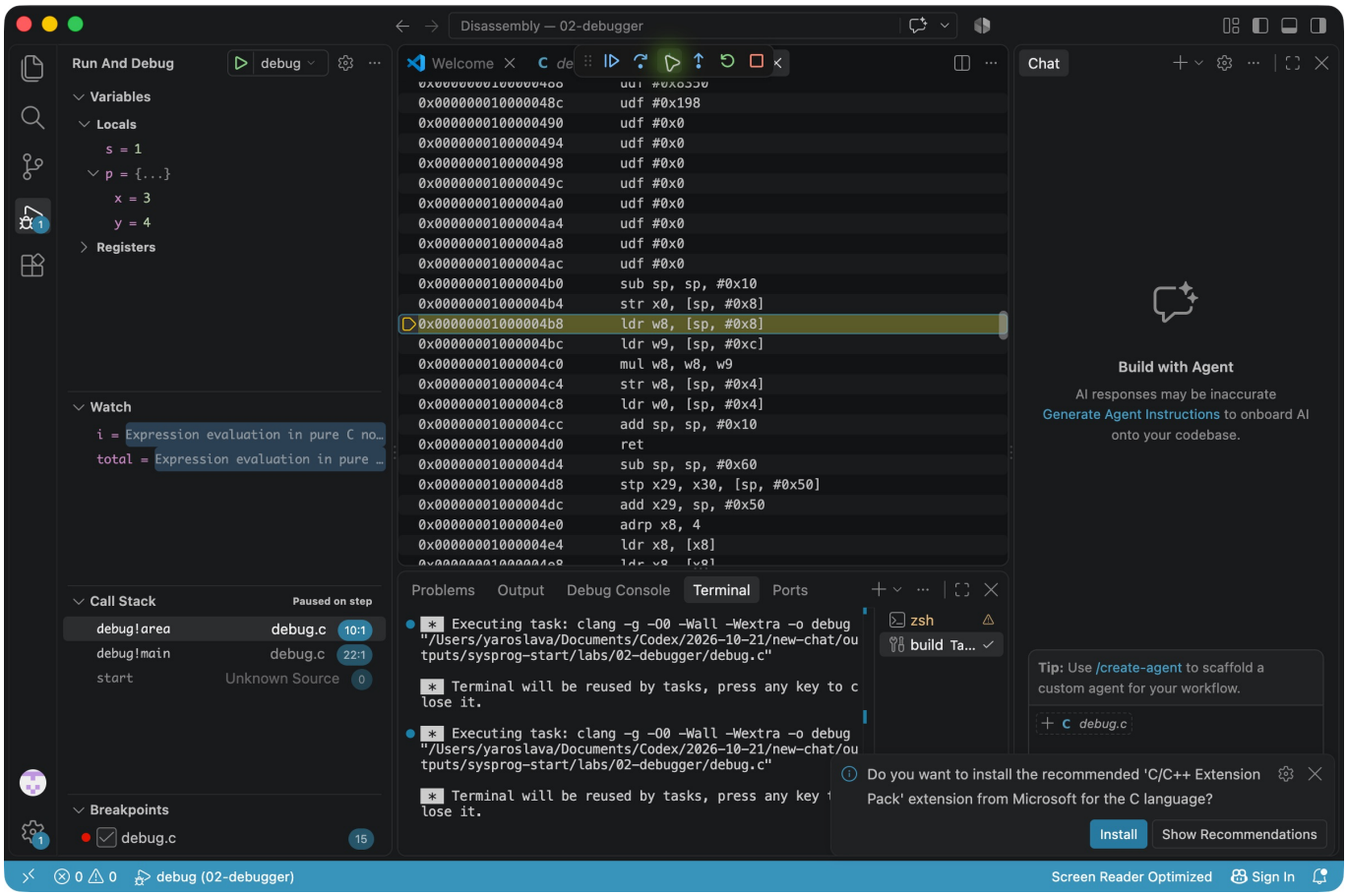
5

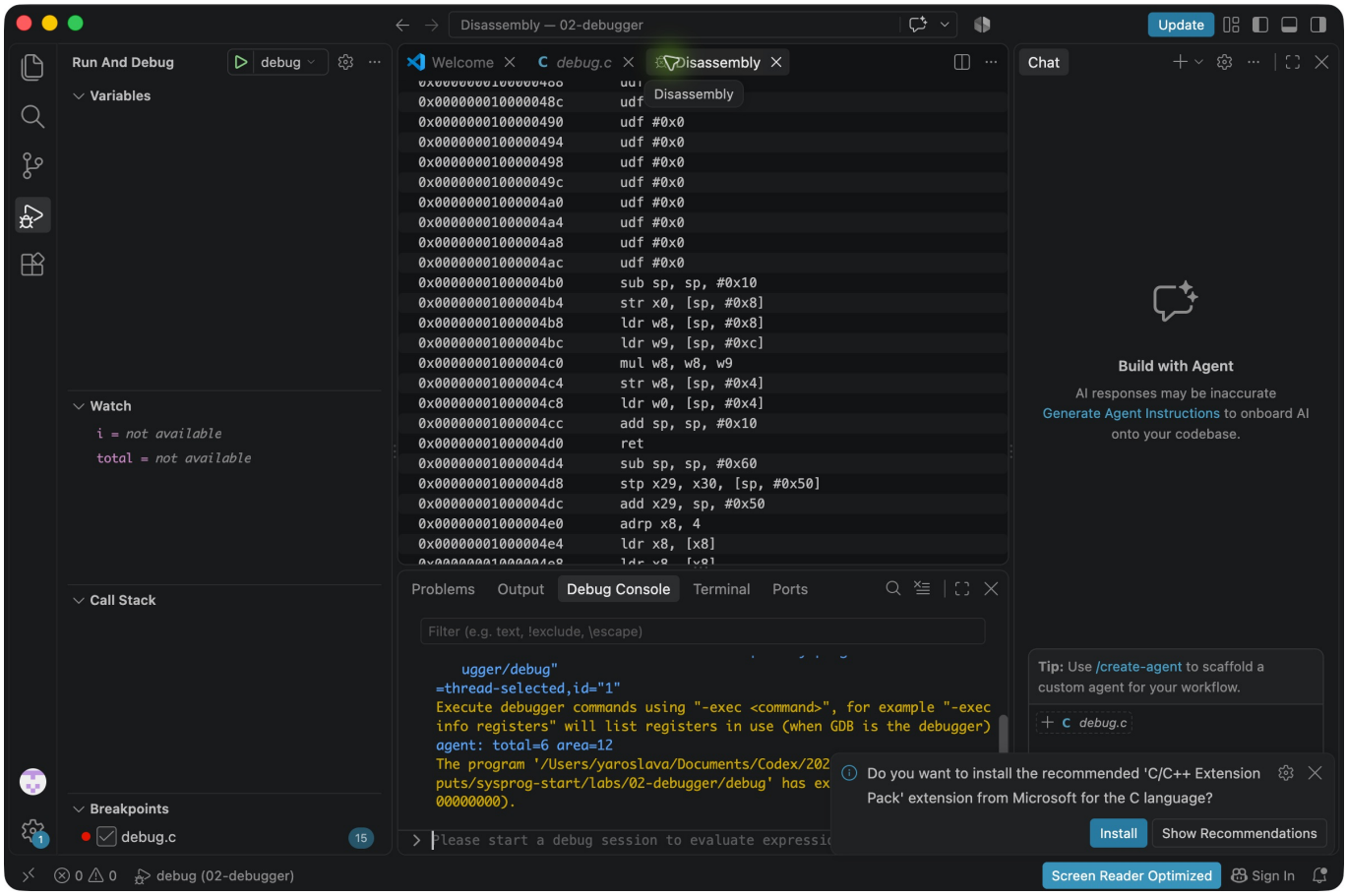

debug.c — 02-debugger
Run And Debug
debug
Variables
Locals
p = {...}
x = 3
y = 4
name = [8]
[0] = 97 'a'
[1] = 103 'g'
[2] = 101 'e'
[3] = 110 'n'
[4] = 116 't'
[5] = 0 '\0'
Watch
i = 2
total = 1
Call Stack
Paused on step
debug!main
debug.c 21:1
start
Unknown Source 0
Breakpoints
debug.c 15
Welcome
debug.c > main(void)
4 struct point {
5 int x;
6 int y;
7 };
8
9 int area(struct point p) {
10 int s = p.x * p.y;
11 return s;
12 }
13
14 int main(void) {
15 struct point p = {3, 4};
16 char name[8];
17 int total = 0;
18
19 strcpy(name, "agent");
20 for (int i = 1; i <= 3; i++)
21 total += i;
22 int a = area(p);
23 printf("%s: total=%d area=%d\n", name, total, a);
24 return 0;
25 }
Problems
Output
Debug Console
Terminal
Ports
Executing task: clang -g -O0 -Wall -Wextra -o debug "/Users/yaroslava/Documents/Codex/2026-10-21/new-chat/ou
tputs/sysprog-start/labs/02-debugger/debug.c"
Terminal will be reused by tasks, press any key to c
lose it.
Executing task: clang -g -O0 -Wall -Wextra -o debug "/Users/yaroslava/Documents/Codex/2026-10-21/new-chat/ou
tputs/sysprog-start/labs/02-debugger/debug.c"
Terminal will be reused by tasks, press any key t
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zsh
build Ta...
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Show Recommendations
0 0
debug (02-debugger)
Screen Reader Optimized
Ln 21, Col 1
Spaces: 4
UTF-8
LF
C
Sign In
Mac

6


debug.c — 02-debugger
Run And Debug
debug
Variables
Locals
s = 1
p = {...}
x = 3
y = 4
Registers
Watch
i = Expression evaluation in pure C no…
total = Expression evaluation in pure …
Call Stack
Paused on step
debug!area debug.c 10:1
debug!main debug.c 22:1
start Unknown Source 0
Breakpoints
debug.c 15
Welcome
debug.c > area(point)
#include <stdio.h>
#include <string.h>
struct point {
int x;
int y;
};
int area(struct point p) {
int s = p.x * p.y;
return s;
}
int main(void) {
struct point p = {3, 4};
char name[8];
int total = 0;
strcpy(name, "agent");
for (int i = 1; i <= 3; i++)
total += i;
int a = area(p);
printf("%s: total=%d area=%d\n", name, total, a);
Problems
Output
Debug Console
Terminal
Ports
zsh
build Ta...
Executing task: clang -g -O0 -Wall -Wextra -o debug "/Users/yaroslava/Documents/Codex/2026-10-21/new-chat/outputs/sysprog-start/labs/02-debugger/debug.c"
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Show Recommendations
0 0
debug (02-debugger)
Screen Reader Optimized
Ln 10, Col 1
Spaces: 4
UTF-8
LF
C
Sign In
Mac

7


Disassembly — 02-debugger
Run And Debug
debug
Variables
Locals
s = 1
p = {...}
x = 3
y = 4
Registers
Watch
i = Expression evaluation in pure C no…
total = Expression evaluation in pure …
Call Stack
Paused on step
debug!area debug.c 10:1
debug!main debug.c 22:1
start Unknown Source 0
Breakpoints
debug.c 15
0x000000010000048c udf #0x198
0x0000000100000490 udf #0x0
0x0000000100000494 udf #0x0
0x0000000100000498 udf #0x0
0x000000010000049c udf #0x0
0x00000001000004a0 udf #0x0
0x00000001000004a4 udf #0x0
0x00000001000004a8 udf #0x0
0x00000001000004ac udf #0x0
0x00000001000004b0 sub sp, sp, #0x10
0x00000001000004b4 str x0, [sp, #0x8]
0x00000001000004b8 ldr w8, [sp, #0x8]
0x00000001000004bc ldr w9, [sp, #0xc]
0x00000001000004c0 mul w8, w8, w9
0x00000001000004c4 str w8, [sp, #0x4]
0x00000001000004c8 ldr w0, [sp, #0x4]
0x00000001000004cc add sp, sp, #0x10
0x00000001000004d0 ret
0x00000001000004d4 sub sp, sp, #0x60
0x00000001000004d8 stp x29, x30, [sp, #0x50]
0x00000001000004dc add x29, sp, #0x50
0x00000001000004e0 adrp x8, 4
0x00000001000004e4 ldr x8, [x8]
Problems
Output
Debug Console
Terminal
Ports
zsh
build Ta...
Executing task: clang -g -O0 -Wall -Wextra -o debug "/Users/yaroslava/Documents/Codex/2026-10-21/new-chat/outputs/sysprog-start/labs/02-debugger/debug.c"
Terminal will be reused by tasks, press any key to close it.
Executing task: clang -g -O0 -Wall -Wextra -o debug "/Users/yaroslava/Documents/Codex/2026-10-21/new-chat/outputs/sysprog-start/labs/02-debugger/debug.c"
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debug.c
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Install
Show Recommendations
0 0
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Disassembly — 02-debugger
Update
Run And Debug
debug
Variables
Watch
i = not available
total = not available
Call Stack
Breakpoints
debug.c
15
Welcome
debug.c
Disassembly
Disassembly
0x0000000100000488 udf
0x000000010000048c udf
0x0000000100000490 udf #0x0
0x0000000100000494 udf #0x0
0x0000000100000498 udf #0x0
0x000000010000049c udf #0x0
0x00000001000004a0 udf #0x0
0x00000001000004a4 udf #0x0
0x00000001000004a8 udf #0x0
0x00000001000004ac udf #0x0
0x00000001000004b0 sub sp, sp, #0x10
0x00000001000004b4 str x0, [sp, #0x8]
0x00000001000004b8 ldr w8, [sp, #0x8]
0x00000001000004bc ldr w9, [sp, #0xc]
0x00000001000004c0 mul w8, w8, w9
0x00000001000004c4 str w8, [sp, #0x4]
0x00000001000004c8 ldr w0, [sp, #0x4]
0x00000001000004cc add sp, sp, #0x10
0x00000001000004d0 ret
0x00000001000004d4 sub sp, sp, #0x60
0x00000001000004d8 stp x29, x30, [sp, #0x50]
0x00000001000004dc add x29, sp, #0x50
0x00000001000004e0 adrp x8, 4
0x00000001000004e4 ldr x8, [x8]
Problems
Output
Debug Console
Terminal
Ports
Filter (e.g. text, !exclude, \escape)
ugger/debug"
=thread-selected,id="1"
Execute debugger commands using "-exec <command>", for example "-exec info registers" will list registers in use (when GDB is the debugger)
agent: total=6 area=12
The program '/Users/yaroslava/Documents/Codex/202
puts/sysprog-start/labs/02-debugger/debug' has ex
00000000).
Please start a debug session to evaluate expressio
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debug (02-debugger)
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